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elements found in Pteridophytes and Gymnosperms. Sieve cells have sieve areas on their lateral walls only. They are not associated with companion cells.

Sieve Tubes

Sieve tubes are long tube like conducting elements in the phloem. These are formed from a series of cells called sieve tube elements. The sieve tube elements are arranged one above the other and form vertical sieve tube. The end wall contains a number of pores and it looks like a sieve. So it is called as sieve plate. The sieve elements show nacreous thickenings on their lateral walls. They may possess simple or compound sieve plates The function of sieve tubes are believed to be controlled by campanion cells.

In mature sieve tube, Nucleus is absent. It contains a lining layer of cytoplasm. A special protein (P. Protein = Phloem Protein) called slime body is seen in it. In mature sieve tubes, the pores in the sieve plate are blocked by a substance called **callose** (callose plug).The conduction of food material takes place through cytoplasmic strands. Sieve tubes occur only in Angiosperms.

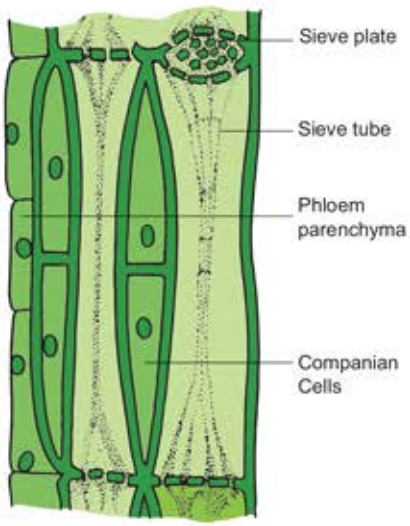


Figure 9.11: Different types of phloem elements

Companion Cells

The thin walled, elongated, specialized parenchyma cells, which are associated with the sieve elements, are called companion cells. These cells are living and they have cytoplasm and a prominent nucleus. They are connected to the sieve tubes through pits found in the lateral walls. Through these pits cytoplasmic connections are maintained between these elements. These cells are helpful in maintaining the pressure gradient in the sieve tubes. Usually the nuclei of the companion cells serve for the nuclei of sieve tubes as they lack them. The companion cells are present only in Angiosperms and absent in Gymnosperms and Pteridophytes. They assist the sieve tubes in the conduction of food materials.

Phloem Parenchyma

The parenchyma cells associated with the phloem are called phloem parenchyma. These are living cells. They store starch and fats. They also contain resins and tannins in some plants. Primary phloem consists of axial parenchyma and secondary phloem consists of both axial and ray parenchyma. They are present in Pteridophytes,Gymnosperms and Dicots.

Phloem Fibres (or) Bast Fibres

The fibres of sclerenchyma associated with phloem are called phloem fibres or bast fibres. They are narrow, vertically elongated cells with very thick walls and a small lumen. Among the four phloem elements, phloem fibres are the only dead tissue. These are the strengthening as well as supporting cells.